



Xander Coetzee

BSC IT & BSC HONS COMPUTER SCIENCE GRADUATE

ABOUT

I am a driven problem-solver with a passion for designing innovative solutions to complex challenges. As a dedicated lifelong learner, I embrace new challenges and opportunities for growth, constantly seeking to expand my skillset and expertise.

EDUCATION

Hoërskool Linden (2017 –2022)

National Senior Certificate

Afrikaans, English, Mathematics, Information Technology, Physical Science, Geography

North-West University (2022 - 2024)

BSc Information Technology

- Computer management and programming
- Designing, development and delivery of computerised systems
- Contribution to management of information and information sources
- Development of IT-related solutions

University of Pretoria (2025)

BSc Honours Computer Science

- Gained advanced technical expertise in Computer Science
- Led and completed a high-standard research project in Digital Forensics
- Applied strong programming and report writing skills across all modules
- Trained in and utilised formal research methodologies.

EXPERIENCE

Assistant lecturer at NWU 2023 – 2024

As an assistant for a programming module, my duties included providing direct coding support to students in labs, grading tests and programming assignments, and managing marks under strict deadlines. I also delivered targeted tutoring on debugging and core concepts. The position required meticulous technical detail and a firm commitment to enhancing student outcomes.

ADDITIONAL ACTIVITIES

- Choral singing in high school and with the NWU PUK Choir has been a joyful way to express my creativity and connect with others through music.
- As a past team leader and manager for an esports team, I enjoyed the camaraderie and the excitement of competitive play, which also fostered a sense of community and collaboration.
- Singing in the university acapella serenade was a fun and unique experience that allowed me to share my passion with a wider audience.
- Playing video games is a favorite pastime that sharpens my problem-solving skills and provides a relaxing escape.

Email

Xandercoetzee1503@gmail.com

Contact no.

+27 82 972 6526

PROJECTS

AI-Driven Phishing & Forensic Readiness

Architected an automated threat detection and evidence preservation framework, integrating secure forensic protocols to ensure organizational cyber-resilience.

Digital Forensics & Information Security

Executed full-lifecycle penetration testing and forensic analysis across Windows/Linux environments, identifying attack vectors and insider threats through advanced disk-level investigation.

Optimised Data Mining Algorithms

Engineered a scalable, memory optimised weighted Apriori algorithm using transaction ID intersections

Computer Vision & Deep Learning

Developed metaheuristic pipelines for automated image quality enhancement and deployed state-of-the-art YOLO models for real-time object detection.

Cross-Lingual NLP & Emotion AI

Pioneered parameter-efficient transfer learning for multilabel emotion analysis, advancing deep learning capabilities for under-resourced African languages.

Skills

- Cyber Security
- Digital Forensics
- Natural Language Processing (NLP)
- Machine Learning & Deep Learning
- Python Programming
- Data Mining
- Research & Academic Writing
- Web Scraping
- Data Visualisation

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REPUBLIC OF SOUTH AFRICA

STATEMENT OF RESULTS / STAAT VAN UITSLAE
NATIONAL SENIOR CERTIFICATE / NASIONALE SENIOR SERTIFIKAAT

NOVEMBER 2021

ALEXANDER GYSBERT COETZEE

EXAMINATION NUMBER / EKSAMENNOMMER: 8211506230025
 ID NUMBER / ID NOMMER 0312155198087
 DATE OF BIRTH / GEBOORTEDATUM 20031215
 CENTRE / SENTRUM: (8150623) HOËRSKOOL LINDEN

(801) GAUTENG GAUTENG DEPARTEMENT VAN ONDERWYS

Code Kode	Subjects Vakke	Percentage Persentasie	Achievement Level Prestasievlak
AFRHL	Afrikaans Huistaal	75	6
ENGFA	Engels Eerste Addisionele Taal	86	7
MATH	Wiskunde	74	6
LIFE	Lewensoriëntering	92	7
GEOG	Geografie	73	6
INFT	Inligtingstegnologie	87	7
PHSC	Fisiese Wetenskappe	65	5
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Die kandidaat kwalifiseer vir die Nasionale Senior Sertifikaat en voldoen aan die minimum vereistes vir baccalaureus graad-, diploma of hoër sertifikaatstudie soos gepubliseer in die Staatskoerant vir toelating tot hoër onderwys.

IK SERTIFISEER DAT DIE OORSPRONKELIKE DOKUMENT WAT AAN MY VIR WAARNEMING VOORGELEI IS, 'N WYSIGING OF VERANDERING OF DIE OORSPRONKELIKE DOKUMENT AANGEBRING IS NIE.

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MAGSNOMMER 72038839
 DATE / DATUM 2022/01/15

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Achievement Level Prestasievlak	Achievement Description Prestasiebeskrywing	Marks / Punte
7	Outstanding achievement / Uitmuntende prestasie	80 - 100
6	Meritorious achievement / Verdienstelike prestasie	70 - 79
5	Substantial achievement / Beduidende prestasie	60 - 69
4	Adequate achievement / Voldoende prestasie	50 - 59
3	Moderate achievement / Matige prestasie	40 - 49
2	Elementary achievement / Basiese prestasie	30 - 39
1	Not achieved / Ontoereikende prestasie	0 - 29

HEAD OF DEPARTMENT
 DEPARTEMENTSHOOF

DIE NASIONALE SENIOR SERTIFIKAAT SAL AAN DIE SKOOL GESTUUR WORD.

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E08P1042 15067

ACADEMIC RECORD

Date: 20 November 2024
Student number: 42167930 - 2022
Title, Name and Surname: MR ALEXANDER GYSBERT COETZEE
Date of Birth: 2003/12/15

Qualification Program: BACHELOR OF SCIENCE IN INFORMATION TECHNOLOGY - 2DXH01
Curriculum: INFORMATION TECHNOLOGY - N302P
Method of Delivery: FULL TIME
Campus: Potchefstroom
Qualification Acceptance Date: 2022/02/01

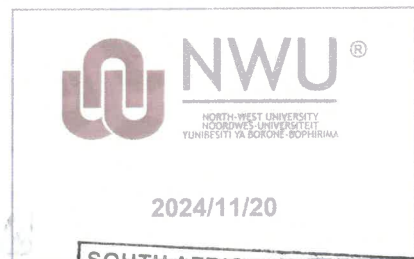
Year	Cr	Module	Module Type	Mark	Result
2022	16	ACCS 1 11 FINANCIAL ACCOUNTANCY (SPECIAL)	Ancillary	84	Distinction
	16	ACCS 1 21 FINANCIAL ACCOUNTING (SPECIAL)	Ancillary	76	Distinction
	12	ALDE 1 11 ACADEMIC LITERACY DEVELOPMENT 1 - ENG	Additional	0	Module Recogni
	12	ALDE 1 22 ACADEMIC LITERACY DEVELOPMENT - ENG	Ancillary	69	Passed
	12	BMAN 1 11 INTRODUCTION TO BUSINESS MANAGEMENT	Ancillary	76	Distinction
	12	CMPG 1 11 INTRODUCTION TO COMPUTING AND PROGRAMMING	Core	81	Distinction
	12	CMPG 1 21 STRUCTURED PROGRAMMING	Core	82	Distinction
	12	CMPG 1 22 USER INTERFACE PROGRAMMING I	Core	82	Distinction
	12	MTHS 1 11 INTRODUCTORY ALGEBRA AND CALCULUS I	Ancillary	42	2nd Op granted
	12	MTHS 1 11 INTRODUCTORY ALGEBRA AND CALCULUS I	Ancillary	52	Passed
	12	STTN 1 11 DESCRIPTIVE STATISTICS	Ancillary	66	Passed
	12	STTN 1 21 INTRO STAT INFERENCE I	Ancillary	55	Passed
2023	16	BMAN 2 23 PROBLEM SOLVING FOR MANAGERS	Ancillary	63	Passed
	16	CMPG 2 11 OBJECT ORIENTED PROGRAMMING	Core	62	Passed
	8	CMPG 2 12 APPS AND ADVANCED USER INTERFACE PROGRAMMING	Core	81	Distinction
	16	CMPG 2 13 SYSTEM ANALYSIS AND DESIGN I	Core	69	Passed
	8	CMPG 2 14 COMMUNICATION SKILLS	Core	67	Passed
	8	CMPG 2 15 INFORMATION SECURITY	Core	60	Passed
	8	CMPG 2 21 DATA STRUCTURES AND ALGORITHMS	Core	66	Passed
	8	CMPG 2 22 DATA ANALYTICS II	Core	62	Passed
	16	CMPG 2 23 SYSTEM ANALYSIS AND DESIGN II	Core	64	Passed
	8	MTHS 2 25 DISCRETE MATHEMATICS	Ancillary	55	Passed
	12	WVNS 2 11 UNDERSTANDING THE NATURAL WORLD	Ancillary	71	Passed
	12	WVNS 2 21 UNDERSTANDING THE NATURAL WORLD	Ancillary	85	Distinction
2024	16	CMPG 3 11 DATABASES	Core	70	Passed
	16	CMPG 3 12 DECISION SUPPORT SYSTEMS 1	Core	69	Passed
	16	CMPG 3 13 ARTIFICIAL INTELLIGENCE	Core	58	Passed
	16	CMPG 3 15 COMPUTER NETWORKS	Core	57	2nd Op granted
	16	CMPG 3 15 COMPUTER NETWORKS	Core	71	Passed
	16	CMPG 3 21 ADVANCED DATABASES	Core	75	Distinction
	16	CMPG 3 22 DECISION SUPPORTS SYSTEMS 2	Core	75	Distinction
	16	CMPG 3 23 IT DEVELOPMENTS	Core	83	Distinction
	16	CMPG 3 24 OPERATING SYSTEMS	Core	72	Passed

Modules for qualification purposes will only be acknowledged if the student meets the necessary admission requirements.

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1 of 2
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FORCE NUMBER
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NAME IN PRINT





Private Bag X6001, Potchefstroom
South Africa, 2520

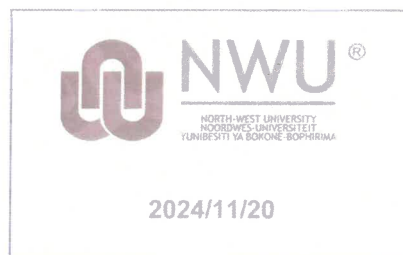
Tel: (018) 299-1111/2222
Web: <http://www.nwu.ac.za>

CERTIFICATE OF CONDUCT

It is hereby certified that the student is/was registered for the abovementioned year/years at the university and that his/her behaviour is/was satisfactory. (In respect of students who are enrolled, this declaration is valid as at the date of issue.) This document is issued without omission or change in any form.

A handwritten signature in black ink, appearing to read 'M. Coetzee', is written over the word 'REGISTRAR'.

REGISTRAR





UNIVERSITEIT VAN PRETORIA
UNIVERSITY OF PRETORIA
YUNIBESITHI YA PRETORIA

Student no: 22708007
Our ref: SL Mgiba
Tel: 0124205316

2025-12-18

This is to certify that

Alexander Gysbert Coetzee
0312155198087

complied with all the requirements for the

Bachelor of Science Honours
in Computer Science

Qualification on 2025-11-28

The qualification will be awarded/conferred at a graduation ceremony to be held on

2026-05-28

for DEAN

Faculty of Engineering, Built Environment and Information Technology



UNIVERSITEIT VAN PRETORIA
UNIVERSITY OF PRETORIA
YUNIBESITHI YA PRETORIA

STUDENT NUMBER 22708007
SURNAME Coetzee
FIRST NAMES Alexander Gysbert
DATE OF BIRTH 2003-12-15
TYPE OF EXEMPTION Admit to Bachelor's Degree

DATE ISSUED 2025-12-18

ACADEMIC RECORD

2025 (Full-time)		Programme: BScHons	Plan: in Computer Science		
Module code	Module name	Credits	%	Results	
COS 760	Natural language processin 760	15	84	Pass with distinction	
COS 783	Digital forensics and inv 783	15	77	Pass with distinction	
COS 781	Data mining 781	15	77	Pass with distinction	
COS 700	Research report 700	30	76	Pass with distinction	
COS 730	Software engineering (I) 730	15	76	Pass with distinction	
COS 791	Special topics (II) 791	15	59	Pass	
COS 720	Computer and info sec (I) 720	15	57	Pass	

Weighted average % for term: 72.75

Cumulative weighted average: 72.75

OUTCOME: Complies with qualification requirements

On 2025-11-28 the abovementioned student complied with all the requirements for the Bachelor of Science Honours qualification. It will be/was awarded/conferred on 2026-05-28.

for DEAN
Faculty of Engineering, Built Environment and Information Technology